

7-1b Relative Price Differences and Purchasing Power Parity

Some countries (such as Switzerland) are famously expensive, and others (such as the Philippines) are known to have cheap prices. How do these price differences affect exchange rates? An answer is provided by the theory of purchasing power parity (PPP), which is essentially the “law of one price.” The theory suggests that in the absence of trade barriers (such as tariffs), the price for identical products sold in different countries must be the same. Otherwise, traders (or arbitragers) may “buy low” and “sell high,” eventually driving prices for identical products to the same level around the world. The PPP theory argues that in the long run, exchange rates should move toward levels that would equalize the prices of an identical basket of goods in any two countries.

One of the most influential and certainly most fun-filled applications of the PPP theory is the Big Mac index, popularized by the *Economist* magazine. The *Economist*’s “basket” is McDonald’s Big Mac hamburger produced in about 120 countries. According to the PPP theory, a Big Mac should cost the same anywhere around the world. In reality, it does not (see [Figure 7.3](#)). In 2014, a Big Mac cost US\$4.8 in the United States and 16.76 yuan in China, which was US\$2.73 according to the nominal exchange rate of 6.14 yuan to the dollar. If the Big Mac indeed cost the same, the de facto exchange rate based on the Big Mac index became 3.49 yuan to the dollar (that is, 16.76 yuan/US\$4.80). According to this calculation, the yuan was 43% “undervalued” against the dollar ($[6.14 - 3.49]/6.14$). But the yuan, which several years ago used to be the most undervalued currency, was only the 12th most-undervalued currency in the Big Mac universe. The Ukrainian hryvnia and the Indian rupee were the most undervalued currencies. In other words, the Big Mac in Ukraine and the chicken-based Maharaja Mac in India (where the beef-based Big Mac is not available) had the best “value” in the world, based on official exchange rates. They only cost US\$1.63 and US\$1.75, respectively.

Figure 7.3

The Big Mac Index

Source: *Economist*, 2014, The Big Mac index: A basket of sliders, July 26: 61. © The Economist Newspaper Limited. Reproduced by permission.

Overall, four observations emerge:

- The Big Mac index confirms that prices in some European countries are very expensive. A Big Mac in Norway was the most expensive in the world, costing US\$7.76.
- Excluding the special case of Ukraine (which was engulfed in a conflict), a Big Mac in South Africa and Indonesia and a Maharaja Mac in India are cheap in dollar terms. This makes sense, because a Big Mac is a product with both traded and non-traded inputs. To simplify our discussion, let us assume that the costs for traded inputs (such as flour for the bun) are the same. It is obvious that non-traded inputs (such as labor and real estate) are cheaper in emerging economies.
- The Big Mac is not a traded product. No large number of American hamburger lovers would travel to South Africa simply to get the best deal on the Big Mac, and then somehow take with them large quantities of the Made-in-South Africa Big Mac (perhaps in portable freezers). If they did that, the Big Mac price in South Africa would

be driven up and the price in the United States would be pushed down—remember supply and demand?

- After having a laugh, we shouldn't read too much into this index. PPP signals where exchange rates may move in the *long run*. But it does not suggest that the yuan should appreciate by 43% or the Norwegian kroner should depreciate by 62% next year. According to the *Economist*, anyone interested in the PPP theory “would be unwise to exclude the Big Mac index from their diet, but Super Size servings (of this index) would equally be a mistake.”

Chapter 7: Dealing with Foreign Exchange: 7-1b Relative Price Differences and Purchasing Power Parity

Book Title: WGU Economics

Printed By: Mike Benton (mbento6@wgu.edu)

© 2021 Cengage Learning, Cengage Learning

© 2026 Cengage Learning Inc. All rights reserved. No part of this work may be reproduced or used in any form or by any means - graphic, electronic, or mechanical, or in any other manner - without the written permission of the copyright holder.